

Model name: 4_preproc_effnetb1retrained_dense128

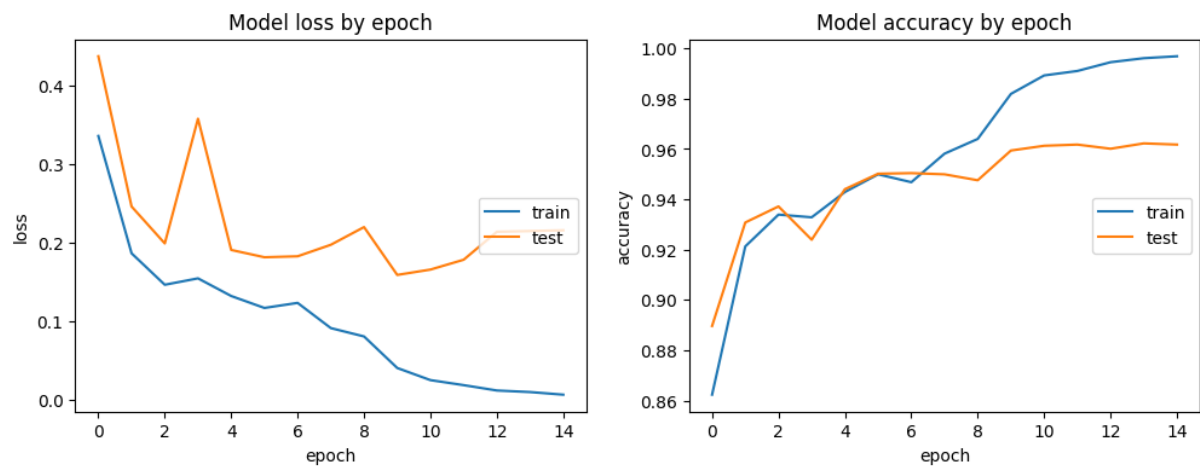
Description and underlying idea:

As another idea of using transfer learning, we used the EfficientNetB1 pretrained model and let it train completely on all layers. The model includes also the EfficientNetB1 preprocessing, so that it can be used on x-ray images directly.

Conclusion:

With an Covid F1 score of 0.99 and average F1 score of 0.96, this model is the best we have found so far. It also performs well when tested on random images from the internet. However, we see that the accuracy quite differs between training and validation dataset, which implies that this model likely has overfitted on the training data.

Training History:



Performance Report / Recall normalized confusion matrix

	pre	rec	f1
COVID	0.994366	0.990182	0.992270
Lung_Opacity	0.944541	0.935622	0.940060
Normal	0.958454	0.964043	0.961240
Viral Pneumonia	0.973244	0.979798	0.976510
avg_pre	0.961712	0.961712	0.961712
avg_rec	0.961729	0.961729	0.961729
avg_spe	0.974701	0.974701	0.974701
avg_f1	0.961709	0.961709	0.961709

